

AI Agents and Agentic AI with Python & Generative AI

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Module 1

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Agentic AI Concepts

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Introduction
Video • 10 min
- ✓

Flipped Interaction Pattern
Video • 7 min
- ✓

The Agent Loop
Video • 8 min
- ✓

Knowledge Check: Agent Loop & Flipped Interaction
Practice Assignment • Grade: 100%
- ✓

Running the Code Samples in the Course
Reading • 10 min
- ✓

Programmatic Prompting for Agents
Ungraded Plugin • 15 min
- ✓

Try Out Programmatic Prompting
Reading • 10 min
- ✓

Programmatic Prompting for Agents II
Ungraded Plugin • 15 min
- ✓

Programmatic Prompting for Agents III
Ungraded Plugin • 15 min
- Programmatic Prompting: Building Blocks for AI Agents
Dialogue • 15 min
- Try Out the Customer Service Agent
Reading • 10 min
- Giving Agents Memory
Ungraded Plugin • 15 min
- Practicing Programmatic Prompting for Agents
Ungraded Plugin • 15 min
- Practicing Programmatic Prompting for Agents (Solution)
Ungraded Plugin • 15 min
- Adding Structure to AI Agent Outputs
Video • 14 min
- Building Your First Agent
Ungraded Plugin • 15 min
- AI Agent / Environment Interface
Ungraded Plugin • 15 min
- AI Agent Feedback and Memory
Ungraded Plugin • 15 min
- The Agent Loop: From Understanding to Thinking Like an Agent
Dialogue • 15 min
- Learning More & Staying Connected
Reading • 10 min
- Understanding Agentic AI Concepts
Graded Assignment • 30 min

Module 2

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AI Agents, Tools, Actions, & Language

Module 3

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GAME: A Conceptual Framework for AI Agents

Module 4

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Agent Tool Mangement

Module 5

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Rethinking How Software is Built in the Age of AI Agents

Programmatic Prompting: Building Blocks for AI Agents

Dialogue

I'm stuck

End Dialogue

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Welcome! In this Dialogue, we'll explore programmatic prompting fundamentals – how to send prompts via code, use message roles effectively, and manage memory and structured information – as the foundation for building autonomous AI agents.

Here's what we'll cover:

- **From Manual Chat to Programmatic Prompting:** Understand the shift from manually typing prompts to programmatically sending prompts using code. This automation is the foundation of the agent loop you learned about.
- **The Power of Message Roles:** Explore how system, user, and assistant roles work together. System sets the ground rules and "programs" agent behavior, user provides input, and assistant represents AI responses for conversation context.
- **Memory Management and Structured Information:** Discover how to control what the LLM knows or remembers between iterations and how to send structured information (like JSON) to enable agents to work with APIs and real-world systems.

Need help? Click "I'm stuck" at the top right to get a hint.

When you're ready, click "Start Dialogue"



✦

Great! Let's dive into **From Manual Chat to Programmatic Prompting**. You've experienced manually typing prompts into a chat interface. How does programmatically sending prompts using code differ from that manual process, and why is this shift crucial for building autonomous AI agents?



S

Sandipan

Manual process uses natural language to communicate with the agent, but the automated process needs an agent to talk to computer using some programming language, making it less much less flexible than humans to interact with

✦

That's a good start! You've identified that programmatic prompting involves an agent interacting with a computer using a programming language, which is correct. However, you mentioned it's "less much less flexible than humans to interact with." Can you elaborate on why you think that might be the case, and how programmatic prompting actually *enables* more autonomous behavior for AI agents, rather than making it less flexible?



S

Sandipan

Use of programming languages makes the agent more autonomous, but at the same time it needs to be able to interact with software in human-like fashion, for example, it may need to fix bugs in code, fix version mismatch etc.

✦

You're touching on a really important point about the challenges and capabilities of AI agents! You're right that using programming languages can make an agent more autonomous. However, the core idea of programmatic prompting is that the *AI agent itself* is generating and sending these prompts, often in natural language, to another language model or system. This is what allows it to act autonomously.

Think about the agent loop you learned about. How does programmatically sending prompts fit into that loop to enable an AI agent to perform tasks without constant human intervention?



Send a message



Dialogue is powered by AI, so check for mistakes and don't share any personal/sensitive information.